Title: Rethinking Forward Secrecy Guarantees: A Zero-knowledge Proof Systems Perspective

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Abstract:

Forward secrecy guarantees remains a critical challenge for symmetric-key primitives, particularly in electronic voting. This paper proposes a framework built on zero-knowledge proof systems that explicitly accounts for protocol formal verification. We evaluate the framework on power traces from an ARM Cortex-M4 platform and compare it against 3 baseline methods. Experimental results show that the proposed approach improves security level in bits by 28.9% while keeping security level in bits within acceptable bounds. We further analyse its limitations and discuss implications for practical deployment.

1. Introduction

The rapid maturation of symmetric-key primitives has changed how researchers and practitioners approach forward secrecy guarantees, yet practical deployment is still limited by scalability, cost, and reproducibility concerns. Over the past decade, zero-knowledge proof systems have attracted sustained attention as a promising route toward solving forward secrecy guarantees in realistic settings such as electronic voting. Despite considerable progress, existing solutions for forward secrecy guarantees tend to degrade when moving from controlled laboratory conditions to production workloads, which motivates a more systematic investigation. A central difficulty is that improvements in key generation latency frequently come at the expense of security level in bits, forcing designers to accept trade-offs that are poorly understood.

Our work is motivated by the growing demand for electronic voting, where reliability and transparency requirements are especially strict. We make three contributions: a unified formulation of forward secrecy guarantees, a practical design built on zero-knowledge proof systems, and an empirical evaluation on power traces from an ARM Cortex-M4 platform. Understanding protocol formal verification is equally important, because it shapes both the theoretical limits and the operational cost of any proposed approach. The remainder of this paper reviews related work, describes the proposed methodology, and concludes with a discussion of limitations and directions for future research.

2. Related Work

Chen et al. (2009) introduced an early framework based on zero-knowledge proof systems, reporting a 30.7% improvement in encryption throughput over previous baselines. Subsequent work by Banerjee et al. (2017) extended this line of research to electronic voting, although the evaluation was restricted to small-scale settings. A survey by Chen et al. (2015) categorised existing approaches to forward secrecy guarantees into three families and highlighted the lack of standard benchmarks. Researchers working on symmetric-key primitives have explored homomorphic encryption libraries as an alternative, with Novak et al. (2009) noting that the approach scales poorly beyond 5000 components. More recently, OConnor et al. (2014) combined zero-knowledge proof systems with homomorphic encryption libraries, achieving a 29.4% reduction in overhead on power traces from an ARM Cortex-M4 platform.

Several studies have criticised the reliance on synthetic workloads, arguing that results on power traces from an ARM Cortex-M4 platform may not generalise to electronic voting. In contrast to these efforts, our work treats forward secrecy guarantees and protocol formal verification jointly rather than as independent concerns. Industry reports corroborate these academic findings, pointing to forward secrecy guarantees as a persistent

obstacle for electronic voting. Despite this body of literature, no prior study has systematically compared zero-knowledge proof systems and homomorphic encryption libraries under identical experimental conditions.

3. Methodology

We formalise forward secrecy guarantees as a constrained optimisation task in which the objective is to maximise security level in bits while keeping encryption throughput within a prescribed budget. The proposed architecture consists of four stages: data acquisition, preprocessing, core processing based on zero-knowledge proof systems, and a validation layer that incorporates homomorphic encryption libraries. To ensure reproducibility, every experiment was repeated 30 times with independent random seeds, and we report mean values together with 95% confidence intervals. Hyperparameters were selected through grid search over a validation split, and the final configuration was fixed before any evaluation on held-out data.

The core routine applies zero-knowledge proof systems to reduce the effect of protocol formal verification, followed by a calibration step that stabilises security level in bits. Our implementation is written in Python and C++, and was executed on a workstation with 64 GB of memory and a single mid-range accelerator. We compare against 3 baseline methods, including a conventional approach built on homomorphic encryption libraries and two recent variants reported in the literature. The proposed design improves security level in bits by 28.9% relative to the strongest baseline and reduces encryption throughput by 24.8%, a statistically significant difference ($p < 0.05$).

Ablation experiments show that removing the homomorphic encryption libraries component reduces security level in bits by approximately 14.6%, confirming its contribution. A sensitivity analysis over 128 parameter settings shows that performance remains stable, with variation in encryption throughput below 9.8%. Threats to validity include the limited diversity of power traces from an ARM Cortex-M4 platform and the possibility that deployment in electronic voting introduces conditions not captured in our setup.

4. Conclusion

This paper investigated forward secrecy guarantees through the lens of zero-knowledge proof systems, and demonstrated that a carefully designed pipeline can improve security level in bits by 28.9% on power traces from an ARM Cortex-M4 platform. The findings suggest that symmetric-key primitives will continue to benefit from closer integration of zero-knowledge proof systems and homomorphic encryption libraries, particularly in electronic voting. Several limitations remain, including the restricted scale of our evaluation and the simplified treatment of protocol formal verification. Future work will extend the framework to larger deployments, explore tighter theoretical guarantees, and release an open benchmark to facilitate fair comparison. We hope the evidence presented here encourages the community to adopt more rigorous and transparent evaluation practices for forward secrecy guarantees.